SURV/SURVMETH 720

Methods to study

Measurement Error

Class #8

November 3, 2025

Today's agenda

1. Midterm make-up exam
2. Continue discussion on validity and reliability
3. Methods to study measurement error

1. Midterm make-up exam

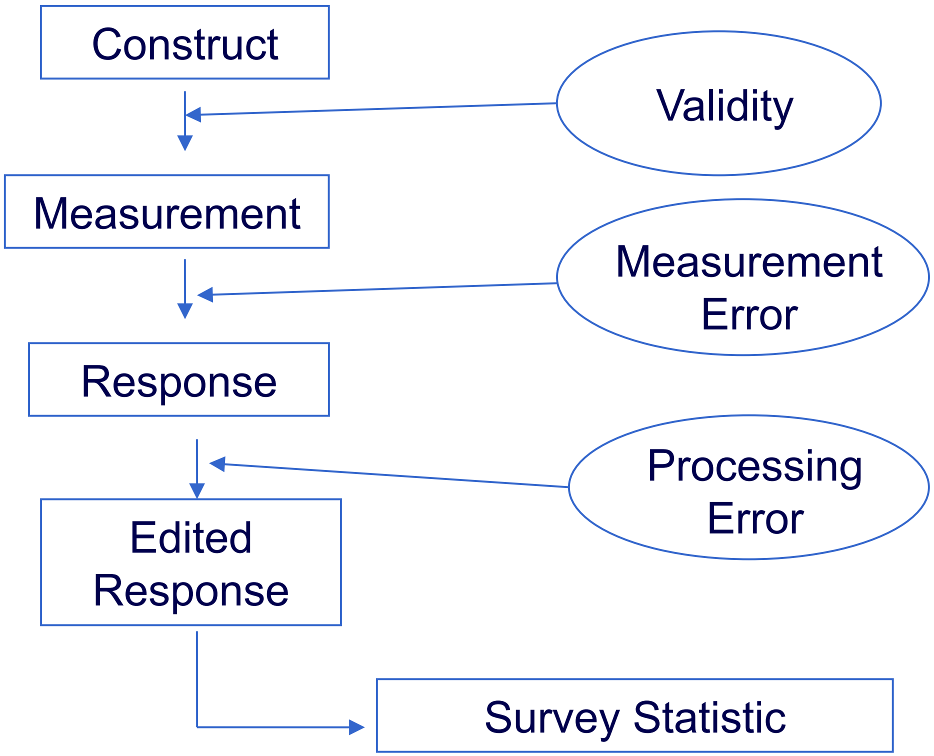
- Optional
  - Those who have done ‘well’ will not be affected by the regrading of others’ exams.
- Take up to 3 questions to rewrite.
- I will regrade these. You get up to 75% of the maximum points of that question (e.g. 7.5 out of 10 points).
- Will open this in Canvas starting tomorrow.
- Deadline: November 12 (2025)

Today's agenda

1. Midterm make-up exam
2. Continue discussion on validity and reliability
3. Methods to study measurement error

Last week….

The survey life cycle: Measurement



- Notion of construct validity
- Measurement error: deviation of the reported value from the "true" value
    - Respondent-level and estimate-level error

Effect of Random Measurement Error

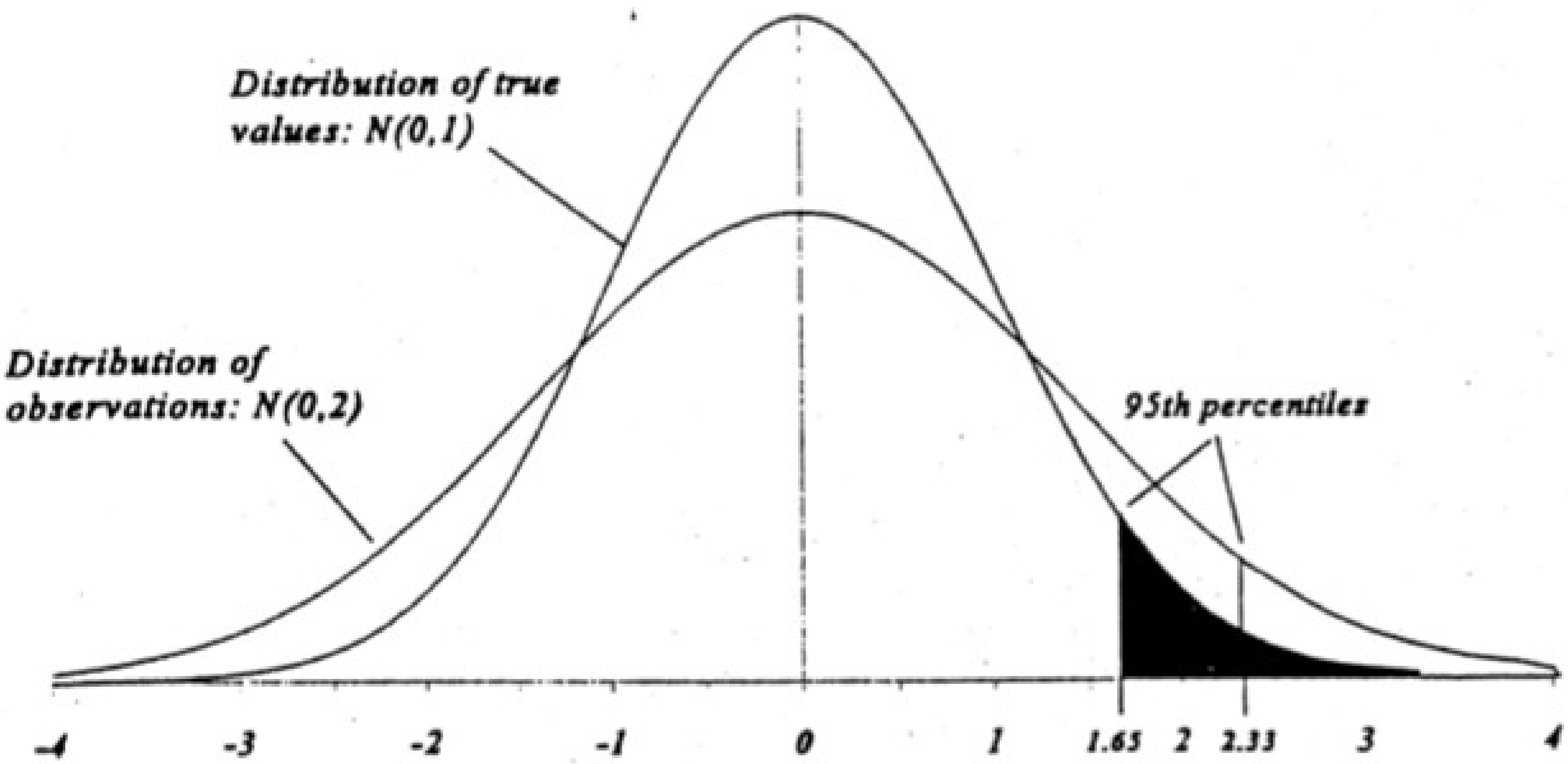


Figure 27.2 The effect of measurement error on the 95th percentile of an N(0, 1) distribution.

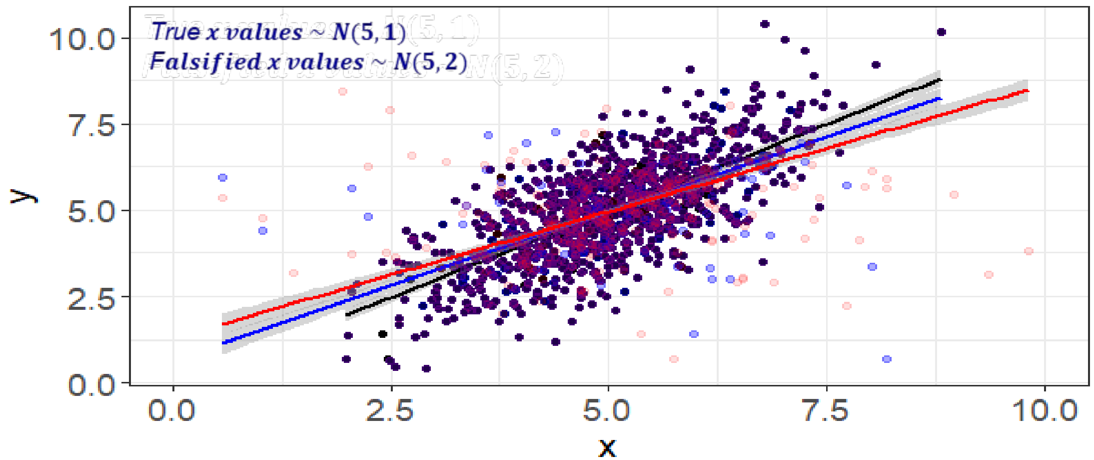
Biemer and Trewin (1997)

True regression line

Regression line (5% falsification)

Regression line (10% falsification)

Simple linear regression



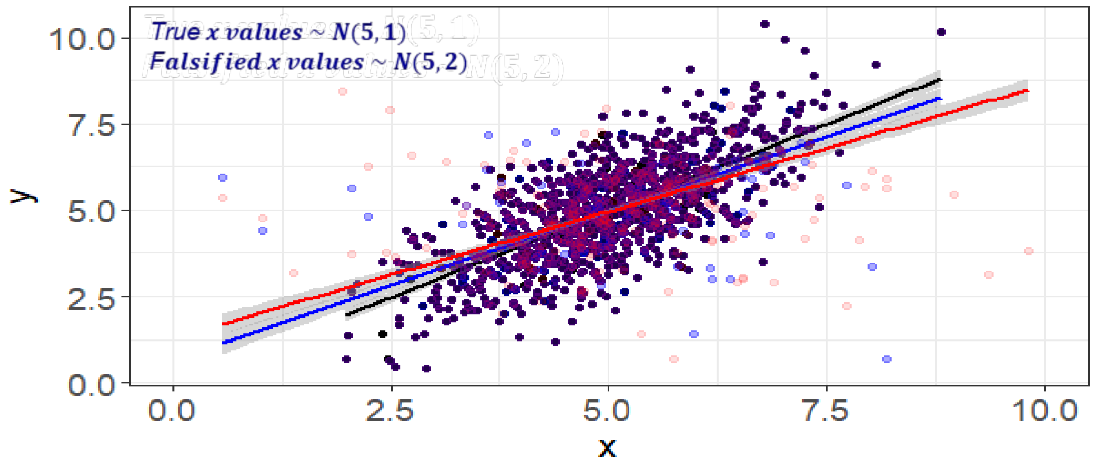
AAPOR-ASA Task force on Falsification
https://www.aapor.org/wp-content/uploads/2022/11/AAPOR_Data_Falsification_Task_Force_Report-updated.pdf

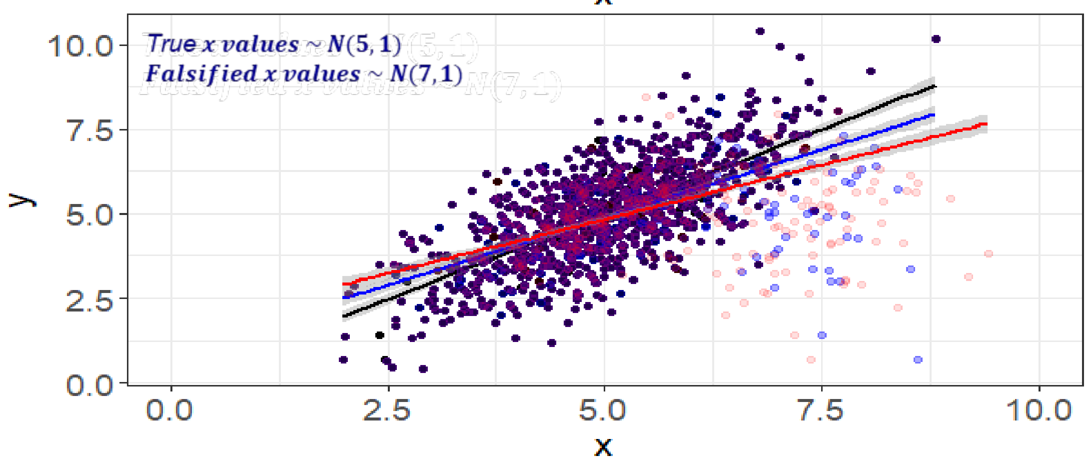
True regression line

Regression line (5% falsification)

Regression line (10% falsification)

Simple linear regression





Falsified x values lead to biased (attenuated) coefficients.

$$\beta_1^{(F)} = \lambda \beta_1^{(T)}$$

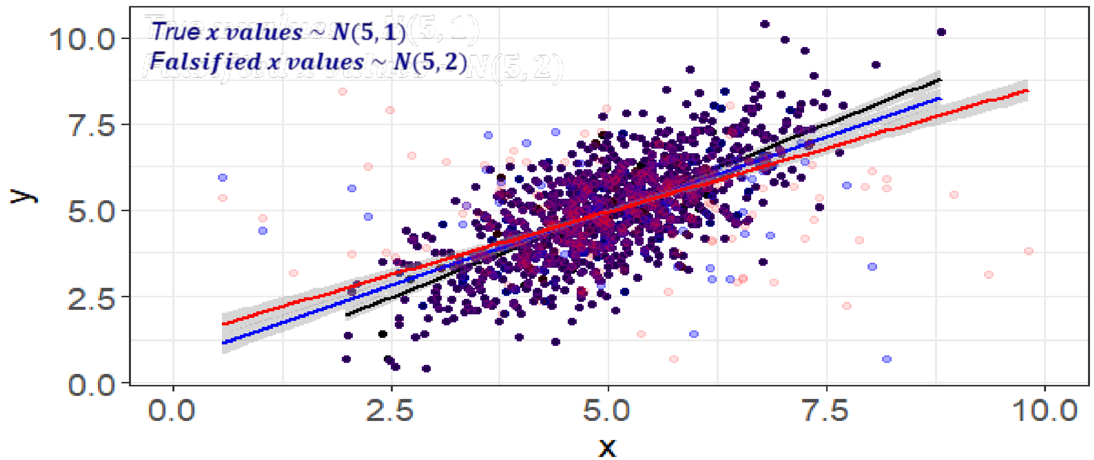
$$\lambda = \frac{\boldsymbol{true\ variance}}{\boldsymbol{total\ variance}}$$

Simple linear regression

True regression line

Regression line (5% falsification)

Regression line (10% falsification)



What will happen to our inferences when there are measurement errors in ‘y’?

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Measurement Error

- Any deviations of the reported value from the “true” value
  - Can be systematic or random
- Has meaning at two levels
  - Respondent level
  - Estimate level
- Two ways of thinking about “truth”
  - Platonic true score
  - Classical true score

Platonic True Score

- $y_{it} = \mu_i + \varepsilon_{it}$

- Construct being measured is at least in theory verifiable based on tangible non-survey evidence
  - In other words, “truth” can be derived from external source
  - Applies to behaviors/individual characteristics
  - Examples: someone’s vote in 2024 election, household size, BMI

- Survey researchers largely adhere to this view

Classical True Score

- Construct being measured has no existing real-world measure
  - Applies to psychological traits and attitudes
  - Applies to latent constructs measured using multi-item scales
  - Examples: extraversion, job satisfaction
- True score is conceptualized as expected value if something were remeasured over a very large number of trials
  - "psychometric perspective"

Classical True Score ...

- At respondent level, a single observation (y_{it}) is the sum of two terms: the mean of the response distribution for an individual, μ_i^*, and an error for the t-th response from it, ε_{it}.

$$y_{it} = \mu_i^* + \varepsilon_{it}$$

Illustration: Classical True Score

- Suppose that on a particular measure of extraversion (0-100), the ith respondent has a score of 52. But on repeated administrations, she achieves scores of 59, 49, and 56.
- What is their 'true' level of extraversion according to the classical score theory?
- What are the response errors for each trial?

Illustration: Measurement Trials …

- True score is the mean of response distribution across trials
  (52+59+49+56)/4 = 54
- Random errors across trails are the deviations from the true score (this is not bias):
  T1: 52-54 = -2
  T2: 59-54 = 5
  T3: 49-54 = -5
  T4: 56-54 = 2

Let’s do a concept review…

1. Validity

- Focusing on "Construct Validity"
  - Conceptual
  - How good are our measures in measuring the construct? How useful/appropriate are the answers to my question in telling me about what I am trying to measure ?
- Consider (person *i*, trial *t*):

$$y_{it} = \mu_i + \epsilon_{it}$$

- Theoretical Validity: Correlation between y and μ (across trials and respondents).

But how to measure this?

- Empirical validity: estimation of theoretical validity
  - Correlation of the measure and another one which is measuring the same construct.

- Lots of terms, variations…

Empirical validity, Criterion validity,
Internal validity, External validity,
Content validity,
Predictive validity, Concurrent validity,
Convergent validity, Divergent validity, etc.

'v/s'

Construct/Theoretical validity

"the principal focus is not on the prediction of a criterion or on the one-to-one correspondence between the content of the test and a specific domain of knowledge or behavior, but on the ability of the test itself to measure the individual trait or characteristic of interest" (Ghiselli et al, 1981)

2. Reliability

- How stable are responses obtained over trials ?
- Operationalized as the ratio of true score variance and observed score variance.
- Reliability and validity are closely linked

How do we square these statements ?

its validity with respect to any criterion measure. It also makes clear what is intuitively obvious: *No measure can be valid without also being reliable, but a reliable measure is not necessarily a valid one*.

Bohrnstedt 1983, 1989, p. 74

bias, on the other. Validity and reliability are both functions of variable response errors made in the measurement process. Given this definition, the traditional statement that "no measure can be valid without also being reliable, but a reliable measure is not necessarily a valid one," is not true. Given the assumptions of classical true score theory, if a measure is reliable, it is valid; if it is valid, it is reliable. Note the contrast with the

Groves, 1989, p. 22

[spreadsheet example]

- From Bohrnstedt, 1983 (last week's reading):

$$\rho_{xy} \leq \rho_{\tau x} = \sqrt{\rho_x}$$

Empirical validity

Theoretical validity

Reliability

- "If a measure is valid it is reliable and if it is reliable it is valid" – this is about **theoretical** validity under the classical true score theory.
- High reliability is a necessary, but not a sufficient, condition for high **empirical** validity
- High **empirical** validity is a necessary and sufficient condition for high reliability.

Psychometric perspective

- Classical true score
- No concept of measurement bias
- Correlations, variances, covariances
- Psychological/Attitudinal
- Validity and Reliability

Survey research perspective

- Platonic true score
- Bias is an important concept
- Correlations, etc. but also means and totals
- Factual, Behavioral
- Bias and Variance

- So, if we could administer a question repeatedly to a respondent, and if the ϵ don't cancel across trials?

  - Psychometric perspective: cannot happen.

$$y_{it} = \mu_i^* + \varepsilon_{it}$$

  - Survey research perspective: bias (our good old dart board analogy)

$$y_{it} = \mu_i + \varepsilon_{it}$$

…and if the ϵ do cancel across trials and persons?

- Psychometric perspective : will anyway happen <u>but</u> variance of scores across trials important (“reliability”)

- Survey research perspective : unbiased mean but additional variance introduced into the estimate. (“simple response variance”)
  - Regression coefficients attenuated when measurement error in the independent variables

- Relationship between validity and bias
  - Closely linked but need not be equivalent.
  - Common usage : “this is a valid estimate” -> “how close to the truth” -> talking more about bias. Requires a true value (classical theory doesn’t recognize this).
  - Also note: What sort of validity are we talking about?
  - A valid measure under the psychometric perspective can still result in a biased estimate for a survey researcher (e.g., a faulty weighing scale). Just be aware of the audience.
- Reliability and “simple response variance” talking of the same thing (but in opposite ways)
- Can you have a reliable measure that is biased ?

- “Do you smoke ?”
  - 15% said “yes” but we found out that an additional 20% smoked but reported “No”.
  - Validity ? Reliability ? Bias ? Variance ?

- “I am very certain about the responses to my question on number of years of formal education. But this is not telling me much about job satisfaction.”
  - Validity ? Reliability ? Bias ? Variance ?

- “A recently conducted survey has yielded highly valid data on binge drinking.”
  - What does this mean ?

II. Methods of studying measurement error

Broadly…

- Different designs for studying bias versus studying variance.

- Researchers of biases often seek cause; those of variances seek estimates of error.

Methods of Studying Measurement Error

1. Question evaluation methods

2. Using measures external to the survey

3. Randomized assignment to different procedures

4. Repeated measures

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Cognitive Interviewing

- Expanded interviews with potential respondents. May include:
  - Think-aloud techniques
  - Definitional/comprehension probes
  - Paraphrasing
  - Vignettes

Are these attempting to study bias or variance ?

Vignette example

"Mrs. Smith is responding to the census questionnaire for her household. She lives with her son, who pays the mortgage, but she doesn't pay anything. How should she respond?"

4. Is this house, apartment, or mobile home — *Mark* ☒ *ONE box*

- ☐ Owned by you or someone in this household with a mortgage or loan? *Include home equity loans.*
- ☐ Owned by you or someone in this household free and clear (without a mortgage or loan)?
- ☐ Rented?
- ☐ Occupied without payment of rent?

Source: Meyers, Mikelyn, Yazmín A. G. Trejo, and Lucia Lykke. 2017. "The Performance of Vignettes in Focus Groups and Cognitive Interviews in a Cross-Cultural Context." *Survey Practice* 10 (3). https://doi.org/10.29115/SP-2017-0017.

Assumptions of Cognitive Interviewing

- Heterogeneity of errors represented in subject pool
  - Participants often recruited from the local community : may not be representative of target population

- Measurement conditions are equivalent to actual survey conditions
  - Attention levels of subjects
  - Motivation in response formation process

Behavior Coding

Involves observing an interview and recording potential correlates of measurement errors

[show IHDS behavior coding]

Paradata

- Data describing the data collection process
  - interviewer observations
  - response latency using time stamps (need a way to capture this)
  - keystroke data indicating changes of responses

Other Question Evaluation Methods

- Expert review
- Formal respondent debriefings
- Focus groups
- Automated methods
  - e.g., Survey Quality Prediction [SQP] https://kodaqs-toolbox.gesis.org/github.com/lukasbirki/tool_sqp/index/
  - Do watch the YouTube tutorial intro: https://bit.ly/3WEgwbM

Challenges

- Some of these are not *directly* addressing measurement error.
- Way to fix problem is not always clear.

In general, methods are complementary

- different methods identify different problems (See Maitland and Presser 2016 https://doi.org/10.1093/jssam/smw014)

Methods of Studying Measurement Error

1. Question evaluation methods

2. Using measures external to the survey

3. Randomized assignment to different procedures

4. Repeated measures

Record Check Study

- Compare self-reports to tangible true value from record data.

  Is this studying bias or variance ?

- Common assumptions:
  - All persons have a record
  - Record data have no errors
  - No errors of matching respondent to record
- Example issues:
  - Income records that are only available for those who file taxes
  - Matching problem if there is be more than one person in precinct voter files with same name

Record Check Studies of Occurrence of Events

Sample from records, obtain self-reports, go back to records and compare

- reverse/backward/retrospective record check

- e.g., Contact patients listed in hospital records, then ask them about their hospital visits.

What do you find striking about this?

Table 7.1 Cases Sampled from Police Records by Whether Crime Was Reported in Survey "Within Past 12 Months" by Type of Crime

Type of Crime	Total Police Cases Interviewed	Percentage Reported to Interviewer as "Within Past 12 Months"
All crimes	394	74.1%
Violent crimes	206	62.6
Assault	81	48.1
Rape	45	66.7
Robbery	80	76.3
Property crimes	188	86.2
Burglary	104	90.3
Larceny	84	81.0

Source: Law Enforcement Assistance Administration, *San Jose Methods Test of Known Crime Victims*, 1972, Table C, p. 6.

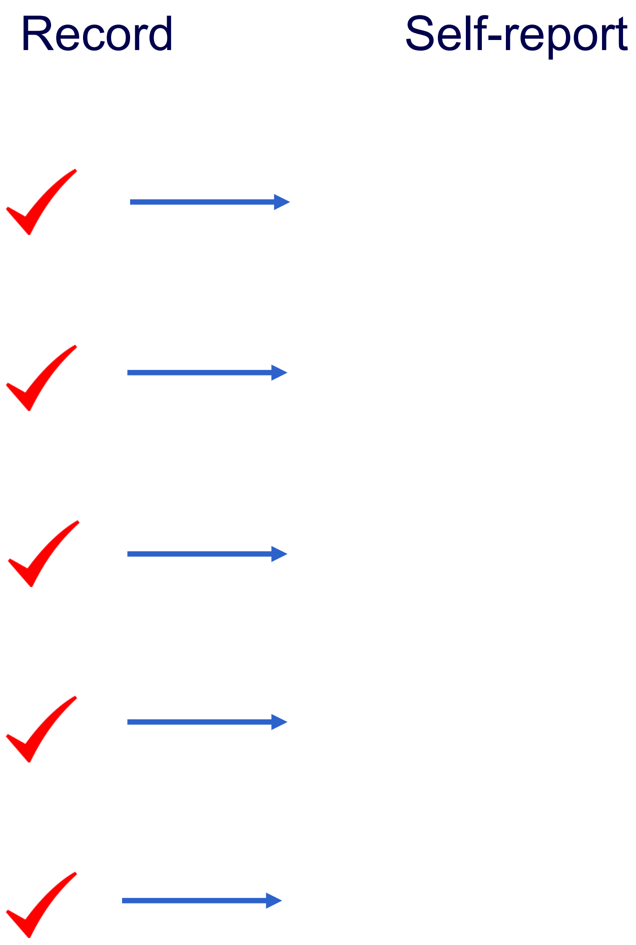
Record Check Studies of Occurrence of Events

- Sample from records, obtain self reports, go back to records and compare (**reverse/backward record check**)
  - Example: Contact patients listed in hospital record, then ask them about their hospital visits.

- Sample using external frame, obtain self-reports and consent to link, confirm with records (forward record check)
  - Example: If hospitalization is reported, then confirm report using hospital records.

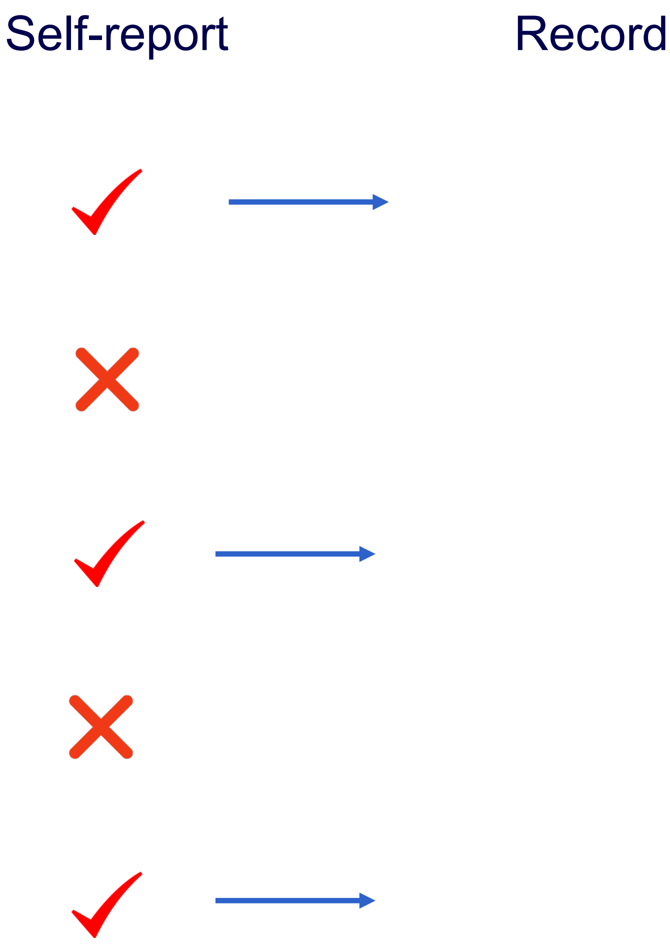
Also have something called “full design record check study” which samples from a frame with all population members and then obtains all records pertaining to the samples: failure to report + erroneous (over) reporting are both dealt with.

Reverse/Backward/Retrospective Record check



Limitation ?

Forward Record check



Limitation?

Limitations of record-check studies

- Assumption that the records have no errors
  - e.g., coverage, measurement
- Mismatch errors
  - Names
  - Events. “I was robbed in May 2023” but no such record exists. Is this an issue of measurement error on date reporting or another event ?
- How generalizable are these studies ?
- Consent to link

3. Use of Aggregate Benchmarks

- Comparing survey estimate to aggregate value based on external data
  - e.g., comparing estimated voting rate in CPS to official voting rate
- Discrepancies often attributed to measurement error, but could be due to other errors (coverage, nonresponse)
- Sub-group analysis ?

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Randomized Treatment

- Randomized experiments (split-ballot, split-sample) within surveys
  - Given a set of respondents, randomized assignment to measurement conditions.
  - Can be used to study both:
    - Measurement error bias (more common) ; think of alternate question wordings; Interest in that **specific** wording. Fixed property. Interest in causes of difference. Reducer.
    - Measurement error variance; think of interviewers – not much interest in specific interviewer/s for inference. Interest in placing qualifications on overall inference. Measurer.
  - Differences between estimates from different treatment groups used to estimate differences in measurement error.

Table 7.2 Percentage of Respondents Forbidding and Not Allowing Speeches in Favor of Communism

Do you think the United States should forbid speeches in favor of communism?		Do you think the United States should allow speeches in favor of communism?	
Yes (forbid)	39.3%	No (not allow)	56.3%
n	(409)	*n*	(432)

Source: H. Schuman and S. Presser, *Questions and Answers in Attitude Surveys*, Academic Press, Orlando, 1981, Table 11.2, p. 281.

- Are the two Q forms measuring the same concept but subject to different response errors? Or are two different concepts being measured?

- What does (56.3 - 39.3) represent?

Contrast this to…

- …a study (also a split-sample) that seeks to study, say, the impact of slower question reading on responses.

  Variable = count of cigarettes smoked in a day.

  Method 1 $\bar{y}$ = 1.8; Method 2 $\bar{y}$ = 2.2

What is the difference with the earlier example ?

Some common indicators of Measurement Bias

- Socially desirable responding
  - Assumption is that procedure that solicits more reports of socially undesirable behaviors or fewer reports of socially desirable behaviors is better
- Satisficing
  - Assumption is that procedure that reduces satisficing behaviors is better (i.e., less missing data, fewer rounded answers, reduced straightlining on grids)

Implicit measurement error model

- Response = True value + Method effect + random error

$$y_{ij} = X_i + M_j + \epsilon_{ij}$$

y_{ij} Survey response by respondent i using method j (e.g. question wording)

X_i True value of characteristic for the ith person

M_j Error associated with jth method (e.g., question delivery – note that the effect can vary by sub-groups)

ε_{ij} Deviation of the ith person from the average effect of the jth method

Potential Limitations of Randomized Designs

- Experimental groups that are not equivalent
  - e.g., IWER group 1 assigned to deliver wording 1 but are generally less successful in recruiting *(also see West and Olson, 2010).*
- Limited external validity
  - If, e.g., IWERs/survey ops aware of the experiment
  - measurement conditions cannot be replicated outside of experiment

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Reinterview Studies

- Involve repeated measurements of same underlying construct at different time points (can also do within the same survey but obvious issues)
  - Preferred method for factual and behavioral questions

What is this studying? Bias or variance?

Intended to *measure* response variance (reliability) rather than *reduce* response bias.

Assumptions (always watch!) of Reinterviews to Measure Response Variance

- No change in true values over trials
  - threatened by long time between trials
- No memory effect of first trial on the response to the second trial
  - threatened by short time between trials
- Other essential survey conditions the same between trials
  - threatened by using supervisors as interviewers

Multi-item scales

- Multi-item scales used as approach for measuring a latent variable (not commonly used for factual and behavior questions)
  - questions are altered each time it is repeated to disguise its redundancy (e.g., *positively worded* and *negatively worded* items)
  - concern: context effects of asking similar questions of the same respondent
- MTMM

Next Week

- Sources of measurement error (interviewer, respondent, questionnaire, mode)
- Reading assignment
- Questions?